Rémy LEROY, PhD

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Engineer and scientific researcher specialised in image processing and computer vision, with a strong intuitive grasp of visual signal interpretation and a unique interdisciplinary profile bridging optics, physics, and deep learning. Skilled in developing maintainable, well-structured code, and driven by a fundamental need to understand underlying mechanisms. Naturally inquisitive, values clarity, simplicity, and reproducibility in algorithm design.

Skill

- Machine Learning
- Computer Vision
- Computational Photography
- Scientific Peer Review
- Clean and Maintainable code
- Python/Pytorch

Work Experience

Post-doctoral researcher @ *INRIA, France* | 2023/04 - 2024/08

Explored **Implicit Neural Representations** techniques for **image compression**, with a focus on Light Field image data.

PhD candidate in signal and image processing @ *ONERA, France* | 2019/03 - 2023/03

Gained deep expertise in state-of-the-art methods for **image processing** and **computer vision**.
Designed and implemented **deep neural networks** using **PyTorch** for **monocular 3D vision**.
Proposed a novel **depth estimation** method that exploits **defocus blur** in single-image inputs.
Developed an end-to-end framework for the **joint optimisation** of optical components and neural networks.
Published and presented original research in peer-reviewed international journals and conferences.

Research intern @ *CEA, France* | 2017/09 - 2018/03

Developed an iterative Kaiser-Squires inversion algorithm in **Python** for removing transformation artefacts in convergence maps for dark matter mass mapping

Education

2023 - Paris-Saclay University, France | **Doctorate in Signal and Image Processing**
2019 - IMT Atlantique (Institut Mines-Telecom), France | **Master's degree in Engineering**
2019 - IMT Atlantique and Rennes 1 University, France | **MSc in Electronic and Automation**

Publications

Multitask deep co-design for extended depth of field and depth from defocus.
M. Dufraisse, R. Leroy, P. Trouvé-Peloux, F. Champagnat, J.-B. Volatier
Proc. **SPIE** 12996, Unconventional Optical Imaging IV, June **2024**

Learning local depth regression from defocus blur by soft-assignment encoding.
R. Leroy, P. Trouvé-Peloux, B. Le Saux, B. Buat et F. Champagnat.
Applied Optics, vol.61, n°29, p.8843--8849. **2022**

Pix2Point: Learning Outdoor 3D Using Sparse Point Clouds and Optimal Transport.
R. Leroy, P. Trouvé-Peloux, F. Champagnat, B. Le Saux, M. Carvlho.
Int. Conf. on **Machine Vision and Applications** (MVA), Oct **2021**